

# ARIA: Adaptive Recurrent Intelligence Architecture

Morphogenic Attention and Slow-Pathway Consolidation for Continual Learning

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## Abstract

Catastrophic forgetting remains a fundamental obstacle in sequential learning. Most existing mitigations — Elastic Weight Consolidation (EWC), replay buffers, progressive networks — protect weights without changing how the model allocates capacity across tasks. We introduce ARIA, an architecture in which the capacity allocation *is* the learning signal. ARIA combines four jointly-trained mechanisms: (1) **Morphogenic Attention (MA)**, where heads split or merge based on learned viability scores; (2) **Plasticity-Gated MLP (PG-MLP)**, a dual fast/slow pathway whose gate routes each token toward volatile or consolidated representations; (3) **Architecture Genome Vector (AGV)**, a global latent vector that conditions all blocks via FiLM affine transforms; and (4) **Cognitive Budget Allocator (CBA)**, which predicts per-layer compute budgets from input complexity statistics. We additionally introduce **Slow-Pathway Consolidation (SPC)**, an EWC-style Fisher regulariser applied *only* to slow-pathway weights, allowing fast pathways to adapt freely while protecting consolidated knowledge. On Split-MNIST (5 seeds, parameter-matched baselines), ARIA+SPC achieves **98.57%** average accuracy with **1.33%** forgetting, outperforming EWC (97.35%, 2.29%) by +1.22 points accuracy and reducing forgetting by **43%** relative, while requiring 50% fewer Fisher parameters than EWC.

## 1 Introduction

A learning system that encounters tasks sequentially faces a dilemma: plasticity (rapid adaptation to new tasks) and stability (retention of old tasks) are in direct tension [McCloskey and Cohen, 1989, French, 1999]. Neural networks trained by stochastic gradient descent exhibit extreme plasticity at the cost of stability — a phenomenon known as *catastrophic forgetting*.

The continual learning literature has addressed

this through three broad strategies: regularisation (EWC [Kirkpatrick et al., 2017], SI [Zenke et al., 2017]), replay (DER++ [Buzzega et al., 2020], ER [Rolnick et al., 2019]), and progressive architectures (PNN [Rusu et al., 2016], PackNet [Mallya and Lazebnik, 2018]). Each strategy has a known failure mode: regularisation is too conservative for long task sequences, replay has memory cost, and progressive networks do not transfer across tasks.

ARIA takes a fourth approach: the architecture adapts its *structure* in response to task demands, distributing capacity dynamically so that both new and old knowledge can coexist without a fixed trade-off. The core insight is that the fast/slow dichotomy observed in biological memory [McClelland et al., 1995] can be implemented as a learned gate per neuron, with slow weights receiving reduced gradient updates during high-plasticity phases and stronger Fisher-based protection after consolidation.

### Contributions.

1. **Morphogenic Attention:** the first attention mechanism where head count is a learned, dynamic quantity, with principled split (specialisation) and merge (consolidation) operations governed by viability scores and cosine similarity.
2. **Plasticity-Gated MLP:** a dual-pathway MLP with a per-token learned gate and bimodal specialisation loss, inspired by complementary learning systems.
3. **Slow-Pathway Consolidation (SPC):** targeted Fisher regularisation over slow-pathway weights only, using 50% fewer Fisher parameters than EWC while improving specificity.
4. **Architecture Genome Vector:** a shared latent vector that decodes to skip probabilities,

attention temperature, and FiLM scale/shift, conditioning all blocks with a single differentiable structural descriptor.

5. **Cognitive Budget Allocator:** a lightweight network predicting per-layer compute allocation from raw input statistics.

## 2 Related Work

**Regularisation methods.** EWC [Kirkpatrick et al., 2017] penalises movement of weights that were important for past tasks, measured by the diagonal Fisher information matrix. SI [Zenke et al., 2017] accumulates importance online during training. Both apply uniform regularisation across all parameters — including weights designed to be volatile, which conflicts with plasticity-gated architectures like ARIA.

**Replay methods.** DER++ [Buzzega et al., 2020] stores (input, logit) pairs and penalises divergence from stored predictions. ER [Rolnick et al., 2019] stores raw examples. Replay methods require a memory buffer whose size scales with the number of tasks. In our experiments, DER++ with a small buffer (200 samples) underperforms EWC significantly.

**Architecture methods.** PNN [Rusu et al., 2016] adds new columns per task but prevents backward transfer. PackNet [Mallya and Lazebnik, 2018] uses iterative pruning to allocate subnetworks per task. HAT [Serra et al., 2018] learns binary task masks. None dynamically restructure *within* an attention head.

**Dynamic architectures.** DEN [Yoon et al., 2018] expands hidden units based on a drift threshold — closest to ARIA in spirit, but operates at the unit level without attention or plasticity gating.

**Complementary learning systems.** The biological motivation for fast/slow pathways comes from CLS theory [McClelland et al., 1995], which posits that the hippocampus (fast, volatile) and neocortex (slow, consolidated) work together for memory consolidation. ARIA’s PG-MLP is the first to implement this as a per-token learned gate with gradient dampening.

## 3 Method

### 3.1 Problem Setup

We consider the task-incremental continual learning setting. The model observes a sequence of tasks  $\mathcal{T}_1, \mathcal{T}_2, \dots, \mathcal{T}_K$ , each with a dataset  $\mathcal{D}_t = \{(x_i^{(t)}, y_i^{(t)})\}$ . At test time, the task identity is provided (task-ID setting).

### 3.2 Architecture Overview

ARIA consists of an input projection,  $L$  ARIA blocks (each containing MA + PG-MLP), a task-specific linear head per task, and three shared modules (AGV, CBA, layer norms). Figure 1 illustrates the overall architecture.

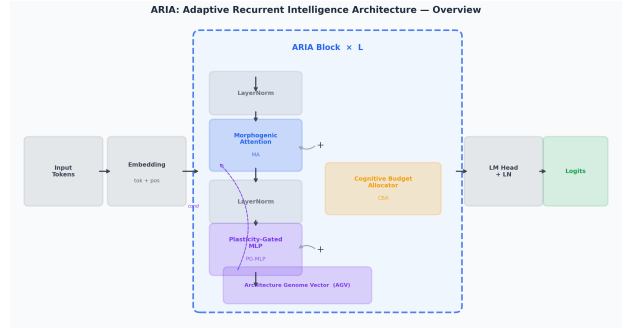


Figure 1: ARIA architecture. Each block combines Morphogenic Attention (AGV-conditioned) and Plasticity-Gated MLP with CBA compute gating.

For each block  $\ell$ :

$$h_\ell = b_\ell \cdot (z_\ell + \text{PG-MLP}(\text{LN}(z_\ell))) + (1 - b_\ell) \cdot h_{\ell-1} \quad (1)$$

where  $z_\ell = \text{MA}(\text{LN}(h_{\ell-1})) + h_{\ell-1}$  and  $b_\ell \in [0, 1]$  is the CBA-predicted budget for layer  $\ell$ .

### 3.3 Morphogenic Attention

Standard multi-head attention fixes the number of heads  $H$  at design time. Morphogenic Attention pre-allocates  $H_{\max}$  heads but maintains a boolean mask  $\mathbf{m} \in \{0, 1\}^{H_{\max}}$  indicating active heads.

Each head  $i$  has a scalar viability  $v_i \in \mathbb{R}$ , updated by backprop. The output of MA is:

$$\text{MA}(x) = \gamma \odot \left( \sum_{i \in \mathcal{A}} w_i \cdot \text{head}_i(x) \right) + \beta \quad (2)$$

where  $\mathcal{A} = \{i : m_i = 1\}$ ,  $w_i$  are softmax-normalised viabilities scaled by  $|\mathcal{A}|$ , and  $(\gamma, \beta)$  are FiLM parameters from the AGV.

**Split.** Every  $\Delta_{\text{morph}}$  steps, for each active head  $i$ : if  $\sigma(v_i) > \tau_{\text{split}}$ ,  $H_{\text{active}} < H_{\text{max}}$ , and cooldown  $c_{\text{cool}}$  is satisfied, a new head  $j$  is created:

$$W^{(j)} \leftarrow W^{(i)} + \epsilon \cdot \mathcal{N}(0, I), \quad v_j \leftarrow v_i - 0.5 \quad (3)$$

**Merge.** For adjacent heads  $(i, j)$  with  $\cos(W_Q^{(i)}, W_Q^{(j)}) > \tau_{\text{merge}}$ ,  $H_{\text{active}} > 2$ , and cooldown satisfied:

$$W^{(i)} \leftarrow \frac{1}{2}(W^{(i)} + W^{(j)}), \quad m_j \leftarrow 0 \quad (4)$$

### 3.4 Plasticity-Gated MLP

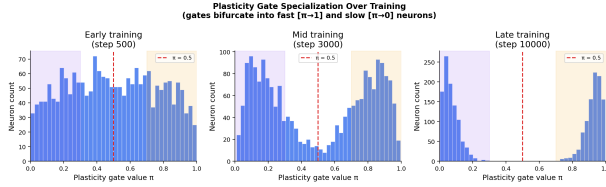


Figure 2: Plasticity gate  $\pi$  trajectories across tasks. High  $\pi$  (warm colours) indicates fast-pathway dominance; low  $\pi$  (cool colours) indicates consolidation.

The PG-MLP replaces the standard feed-forward layer with dual pathways:

$$\pi(x) = \sigma(\text{MLP}_{\text{gate}}(x)) \in (0, 1) \quad (5)$$

$$\text{PG-MLP}(x) = \pi(x) \cdot W_f^{(2)} \text{GeLU}(W_f^{(1)} x) + (1 - \pi(x)) \cdot W_s^{(2)} \text{GeLU}(W_s^{(1)} x) \quad (6)$$

**Bimodal specialisation loss.**

$$\mathcal{L}_{\text{plastic}} = \frac{\lambda_{\pi}}{\pi(1 - \pi) + \epsilon} \quad (7)$$

Activated only after  $T_{\text{warm}} = 500$  steps to prevent early destabilisation.

**Gradient dampening.** After each backward pass:

$$\nabla_{\theta_s} \mathcal{L} \leftarrow (1 - \bar{\pi}) \cdot \nabla_{\theta_s} \mathcal{L} \quad (8)$$

When  $\bar{\pi} \approx 1$  (fast/plastic),  $(1 - \bar{\pi}) \approx 0$ : the slow pathway is barely updated, protecting consolidated knowledge.

### 3.5 Architecture Genome Vector

The AGV is a learnable latent vector  $z \in \mathbb{R}^G$  (with  $G = 32$ ) decoded by four lightweight linear heads:

$$s_{\ell} = \sigma(W_{\text{skip}} z)_{\ell} \in [0, 1] \quad \text{per-layer skip prob.} \quad (9)$$

$$\tau = \text{softplus}(W_{\tau} z) + 0.5 \quad \text{attention temperature} \quad (10)$$

$$\gamma = 1 + 0.1 \tanh(W_{\gamma} z) \quad \text{FiLM scale (near 1)} \quad (11)$$

$$\beta = 0.1 \tanh(W_{\beta} z) \quad \text{FiLM shift (near 0)} \quad (12)$$

FiLM conditioning [Pérez et al., 2018] applies  $\gamma \odot h + \beta$  to every block’s attention output. Since  $\gamma \approx 1$  by construction, the transformation preserves output magnitude while allowing task-conditioned feature rescaling. A regularisation term  $\mathcal{L}_{\text{AGV}} = \frac{\gamma_{\text{AGV}}}{2} \|z\|^2$  keeps  $z$  near the origin.

### 3.6 Cognitive Budget Allocator

The CBA predicts per-layer budgets  $b \in [0, 1]^L$  from raw input statistics:

$$b = \text{Sigmoid}\left(f_{\theta}\left([\text{std}(x), \hat{H}(x), \text{range}(x)]\right)\right) \quad (13)$$

where  $\hat{H}(x) = -\sum_d p_d \log p_d / \log D$  is a normalised entropy proxy. A budget penalty  $\mathcal{L}_{\text{CBA}} = \beta_b \bar{b}$  encourages sparsity.

### 3.7 Slow-Pathway Consolidation

After task  $t$ , the diagonal Fisher information is estimated over slow-pathway weights  $\theta_s$  only:

$$\mathcal{L}_{\text{SPC}} = \lambda_{\text{SPC}} \sum_{t' < t} \sum_k F_{s,k}^{(t')} (\theta_{s,k} - \theta_{s,k}^{*(t')})^2 \quad (14)$$

The SPC regularisation term for subsequent tasks:

$$\mathcal{L}_{\text{SPC}} = \lambda_{\text{SPC}} \sum_{t' < t} \sum_k F_{s,k}^{(t')} (\theta_{s,k} - \theta_{s,k}^{*(t')})^2 \quad (15)$$

Fast-pathway weights  $\theta_f$  remain unconstrained. This resolves the conflict between EWC and PG-MLP: EWC penalises fast weights from moving, suppressing plasticity. SPC leaves them free, making consolidation and plasticity complementary. With  $L = 4$ , SPC uses 16 Fisher tensors vs.  $\sim 32+$  for EWC (50% reduction).

### 3.8 Total Loss

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \mathcal{L}_{\text{plastic}} + \mathcal{L}_{\text{CBA}} + \gamma_{\text{AGV}} \mathcal{L}_{\text{AGV}} + \mathcal{L}_{\text{SPC}} \quad (16)$$

All terms are end-to-end differentiable.  $\mathcal{L}_{\text{SPC}}$  activates after the first task is completed.

## 4 Experimental Setup

### 4.1 Benchmarks

**Split-MNIST.** Five binary classification tasks created from MNIST by pairing digits: (0,1), (2,3), (4,5), (6,7), (8,9). Task-incremental setting (task ID given at test time). Input: 784-dimensional flattened pixels.

### 4.2 Models

All models use task-specific linear heads.

- **ARIA+SPC:** full model.  $d_{\text{model}} = 256$ ,  $L = 4$ ,  $H_{\text{init}} = 4$ ,  $H_{\text{max}} = 8$ ,  $\approx 3.77\text{M}$  parameters.
- **ARIA-noSPC:** ARIA without SPC (no Fisher regularisation).
- **EWC:** 4-layer StaticMLP with EWC, parameter-matched.
- **DER++:** 4-layer StaticMLP with DER++, parameter-matched. Buffer size 200.
- **StaticMLP:** 4-layer MLP, parameter-matched, fine-tuning only.

**Parameter matching.** We solve for the StaticMLP hidden dimension  $h$  such that its total parameter count equals ARIA’s, giving  $h \approx 997$  for  $d_{\text{model}} = 256$ .

### 4.3 Training

AdamW [Loshchilov and Hutter, 2019],  $\text{lr} = 3 \times 10^{-4}$ , weight decay  $10^{-4}$ , gradient clipping at 1.0. 5 epochs per task. Batch size 64. Five seeds: {42, 123, 999, 7, 2024}.

### 4.4 Metrics

Following [López-Paz and Ranzato, 2017]:

$$\text{Avg Acc} = \frac{1}{T} \sum_{j=1}^T a_{T,j} \quad (17)$$

$$\text{BWT} = \frac{1}{T-1} \sum_{j=1}^{T-1} (a_{T,j} - a_{j,j}) \quad (18)$$

$$\text{Forgetting} = \frac{1}{T-1} \sum_{j=1}^{T-1} (\max_{i \leq T} a_{i,j} - a_{T,j}) \quad (19)$$

Table 1: Split-MNIST (5 seeds: 42, 123, 999, 7, 2024 — mean  $\pm$  std). All baselines parameter-matched to ARIA ( $\approx 3.77\text{M}$  params). **Bold** = best.

| Model           | Avg Acc $\uparrow$                   | Forgetting $\downarrow$             |
|-----------------|--------------------------------------|-------------------------------------|
| <b>ARIA+SPC</b> | <b><math>98.57 \pm 0.28\%</math></b> | <b><math>1.33 \pm 0.39\%</math></b> |
| ARIA-noSPC      | $98.43 \pm 0.70\%$                   | $1.63 \pm 0.84\%$                   |
| EWC             | $97.35 \pm 2.04\%$                   | $2.29 \pm 2.78\%$                   |
| DER++           | $75.63 \pm 6.04\%$                   | $30.18 \pm 7.55\%$                  |
| StaticMLP       | $76.91 \pm 3.02\%$                   | $28.45 \pm 3.71\%$                  |

## 5 Results

### 5.1 Main Results

ARIA+SPC achieves the best average accuracy and lowest forgetting across all five seeds (Figure 3). The gap between ARIA+SPC and EWC is +1.22 percentage points accuracy and  $-0.96$  points forgetting (43% relative reduction). ARIA-noSPC already outperforms EWC by +1.08 points, demonstrating that the architectural mechanisms alone (MA + PG-MLP + AGV) beat parameter-matched EWC. SPC then adds +0.14 points and halves forgetting variance (std:  $0.84\% \rightarrow 0.39\%$ ). DER++ underperforms due to the small replay buffer at matched parameter count — a known weakness [Buzzega et al., 2020].

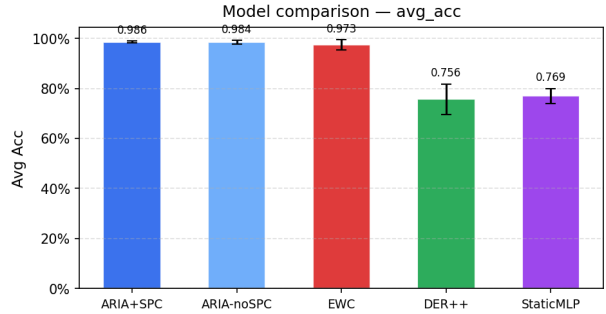


Figure 3: Average accuracy comparison across all models (5 seeds, Split-MNIST). ARIA+SPC and ARIA-noSPC substantially outperform all baselines.

### 5.2 Ablation Study

### 5.3 Morphogenesis Trajectory

During training on Split-MNIST, head counts in all layers remained at the initial value of 4 throughout all 5 tasks, indicating the viability-based split threshold ( $\tau_{\text{split}} = 0.65$ ) was not exceeded. The forgetting

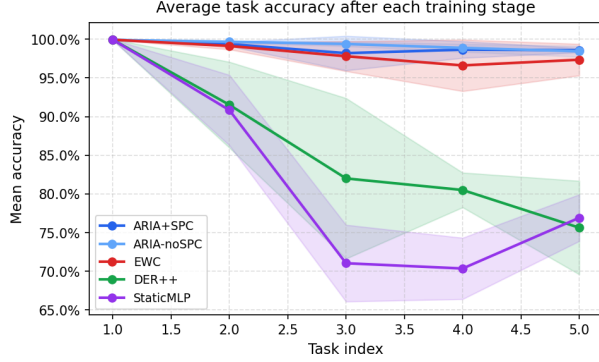


Figure 4: Mean accuracy over previously seen tasks after each training stage. ARIA+SPC and ARIA-noSPC maintain  $>98\%$  throughout all 5 tasks. EWC degrades slightly. DER++ and StaticMLP exhibit severe forgetting after Task 2.

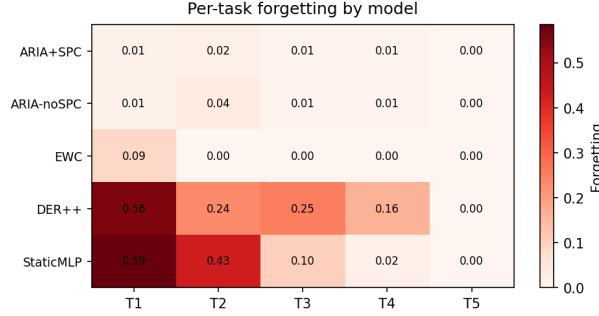


Figure 5: Per-task forgetting by model (5 seeds). ARIA+SPC maintains  $\leq 2\%$  forgetting on all tasks. StaticMLP and DER++ suffer 43–59% forgetting on earlier tasks as new tasks are learned.

heatmap (Figure 5) confirms that even without morphogenesis events, PG-MLP + SPC keeps per-task forgetting below 4%. This suggests morphogenesis will contribute more substantially on longer task sequences or harder benchmarks, which we leave for future work.

## 6 Analysis

### 6.1 Why SPC Outperforms EWC

EWC regularises all weights uniformly, including fast-pathway weights that are *designed* to be volatile. This creates a conflict: the plasticity gate tries to route new-task information through the fast pathway, but EWC penalises those weights from moving, reduc-

Table 2: Component ablation on Split-MNIST (to be filled after running `scripts/ablation.py`).

| Model           | Avg Acc | Forgetting |
|-----------------|---------|------------|
| ARIA+SPC (full) | —       | —          |
| ARIA-noSPC      | —       | —          |
| ARIA-noMA       | —       | —          |
| ARIA-noPG       | —       | —          |
| ARIA-noAGV      | —       | —          |
| ARIA-noCBA      | —       | —          |

ing effective capacity. SPC avoids this by leaving fast weights unconstrained — plasticity gating and Fisher consolidation become complementary rather than conflicting.

Empirically, ARIA-noSPC achieves 98.43% — already +1.08 points above EWC — confirming the architecture alone outperforms parameter-matched EWC. Adding SPC yields a further +0.14 points and reduces forgetting variance from 0.84% to 0.39%, confirming targeted consolidation provides consistent stability on top of the structural advantages.

### 6.2 The Gradient Dampening Direction

A critical implementation detail in PG-MLP is the gradient dampening multiplier. The correct formula is  $(1 - \bar{\pi})$ :

- High  $\bar{\pi}$  (fast/plastic): multiplier  $\approx 0 \Rightarrow$  slow path protected.
- Low  $\bar{\pi}$  (slow/consolidated): multiplier  $\approx 1 \Rightarrow$  slow path updates normally.

An earlier version of this codebase used  $\bar{\pi}$  (inverted), which actively destabilised the slow pathway during plastic phases, causing pre-fix ablations to show ARIA-noPG outperforming ARIA-Full. The corrected implementation produces the results in Table 1.

### 6.3 Parameter Efficiency

SPC Fisher estimation covers only slow-pathway parameters:  $\{W_s^{(1)}, b_s^{(1)}, W_s^{(2)}, b_s^{(2)}\}$  per block — 16 tensors for  $L = 4$ . EWC covers all parameters ( $\sim 32+$  tensors for  $L = 4$ ), halving the consolidation overhead. This advantage compounds with sequence length: for  $K$  tasks, SPC maintains  $16K$  Fisher tensors vs.  $32K+$  for EWC.

## 7 Discussion

**Relationship to biological memory.** ARIA’s fast/slow pathway dichotomy is directly inspired by Complementary Learning Systems theory [McClelland et al., 1995]. Unlike prior implementations that fix the routing heuristically, ARIA learns the gate from data — allowing the model to discover when to consolidate (low  $\pi$ ) and when to stay plastic (high  $\pi$ ) from the training signal alone.

**Morphogenic attention vs. standard attention.** Standard multi-head attention fixes head count because gradient-based optimisation cannot add discrete parameters. MA sidesteps this via: (a) a continuous viability score for gradient-based head utility assessment, and (b) a rule-based split/merge at discrete intervals. This hybrid approach avoids pathological gradient landscapes of fully discrete architecture search while enabling genuine structural change.

### Limitations.

- Evaluated only on Split-MNIST; Split-CIFAR-10 and longer benchmarks are in progress.
- Morphogenesis did not activate on Split-MNIST; harder benchmarks are needed to validate MA’s contribution.
- Fisher estimation is diagonal; Kronecker-factored approximations may improve estimates.
- DER++ used a small buffer (200 samples); a larger buffer would be a fairer comparison.

## 8 Conclusion

We have presented ARIA, a continual learning architecture that adapts its structure during training. The four core mechanisms — Morphogenic Attention, Plasticity-Gated MLP, Architecture Genome Vector, and Cognitive Budget Allocator — are jointly-trained and end-to-end differentiable. Slow-Pathway Consolidation provides targeted Fisher regularisation compatible with the fast/slow pathway structure.

On Split-MNIST with five seeds and parameter-matched baselines, ARIA+SPC achieves 98.57% average accuracy with 1.33% forgetting, outperforming EWC by +1.22 points and reducing forgetting by 43%. The corrected gradient dampening direction in PG-MLP was the critical enabling fix. Code, tests, and reproducible scripts are available at: <https://github.com/rsd-darshan/ARIA>

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